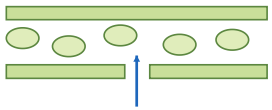
Plant tissues & transport

Structure → function

1 Leaf tissues



CO_2 enters through a stoma

Epidermis/cuticle protect and limit evaporation. Palisade cells absorb light using chloroplasts.

Spongy mesophyll air spaces shorten gas paths. Guard cells control stomata.

2 Roots & growing tips

Root hairs increase surface area: water enters by **osmosis,** minerals by **active transport.**

Meristems at root/shoot tips contain dividing cells that differentiate.

Move substances

3 Xylem & phloem

Xylem: dead hollow tubes, lignin support; root → leaf water/minerals.

Phloem: living elongated cells with end-wall pores; dissolved sugars move by **translocation** to growing/storage tissues.

4 Transpiration

Water evaporates from mesophyll; vapour diffuses out through stomata.

Water loss helps pull a continuous stream up xylem. Guard cells balance gas exchange and water loss.

Measure water loss

5 Changing the rate

Heat → faster evaporation/diffusion. Wind removes humid air → steeper gradient.

Humidity → smaller gradient, slower loss. Light usually opens stomata, increasing loss if water is available.

6 Potometers & sampling

Cut shoot under water; airtight setup; dry leaves; acclimatise. Change one factor, control others; repeat.

Measures **uptake,** not exact loss. Rate = distance/time; volume = **πr^2 × distance.**

Count stomata in several random measured areas; report mean number per area.